

AMAZON SALES ANALYSIS DASHBOARD

Power BI Project Documentation

End-to-end sales performance reporting built on Power BI

1. Project Overview

This project is an end-to-end Power BI dashboard built to analyze Amazon product sales performance. It covers the full analytics workflow — importing and cleaning raw sales data from a CSV source, modeling the data, building DAX measures for time-intelligence and KPI calculations, and designing an interactive report with charts, cards, and slicers for business users to explore sales trends, top products, and category performance.

Tool Used	Microsoft Power BI (Power Query, Data Model, DAX, Report View)
Data Source	Amazon product sales data (CSV file)
Project Type	Sales performance / retail analytics dashboard
Core Focus	YTD & QTD sales tracking, product performance, and customer review analysis

2. Problem Statement

The business needed a single, interactive view of Amazon product sales to monitor revenue performance, spot trends, and understand which products and categories drive the most sales and customer engagement. The dashboard was designed to answer the following KPI requirements:

- **YTD Sales:** Monitor year-to-date sales to gauge overall revenue performance over time.
- **QTD Sales:** Track quarter-to-date sales figures to identify sales trends and fluctuations.
- **YTD Products Sold:** Analyze the total number of products sold throughout the year to understand product movement.
- **YTD Reviews:** Keep tabs on year-to-date product reviews to assess customer feedback and satisfaction.

3. Chart & Visualization Requirements

The report was designed around the following visuals, each mapped to a specific analytical need:

- **Sales by Month (Line Chart):** Visualizes sales trends over time on a monthly basis to identify seasonal patterns and growth trends.
- **Sales by Week (Column Chart):** Displays sales data on a weekly basis to pinpoint shorter-term fluctuations and performance insights.
- **Sales by Product Category (Table / Heat Map):** Provides a high-level, color-coded overview of sales across different product categories.
- **Top 5 Products by YTD Sales (Bar Chart):** Highlights the top-performing products based on year-to-date sales to focus on key revenue generators.
- **Top 5 Products by YTD Reviews (Bar Chart):** Identifies the top-rated products by year-to-date reviews to understand customer preferences.

4. Functionalities & Skills Applied

Building this dashboard involved the full Power BI toolset, from raw data import through to polished, interactive visuals:

Data & Modelling Skills	DAX, Formatting & Design Skills
Connecting to CSV File	Calculate
Data Cleaning	YTD Functions
Data Modelling	QTD Functions
Data Processing	Custom Sorting
Power Query	Formatting
Date Tables	Conditional Formatting
Time Intelligence Functions	New Card Visual
DAX	Creating Charts
Text Function	Formatting Visuals
Date Function	Creating Functions
Filter Function	Navigations

5. Data Preparation & Modelling

Sales data was imported from a CSV source and cleaned/shaped using Power Query. Key steps included:

- Connecting to the raw Amazon sales CSV file and loading it into Power BI.
- Cleaning inconsistent data — removing errors, fixing data types, and standardizing product and category fields.
- Building a dedicated Date table to support time-intelligence calculations (YTD, QTD, weekly and monthly trends).
- Establishing relationships between the sales data and the date table within the data model.
- Writing DAX measures for YTD Sales, QTD Sales, YTD Products Sold, and YTD Reviews using time-intelligence and filter functions.

6. Final Dashboard

The completed report brings together all KPI cards, trend charts, category breakdown, and top-product rankings into a single, dark-themed, interactive Amazon sales dashboard:

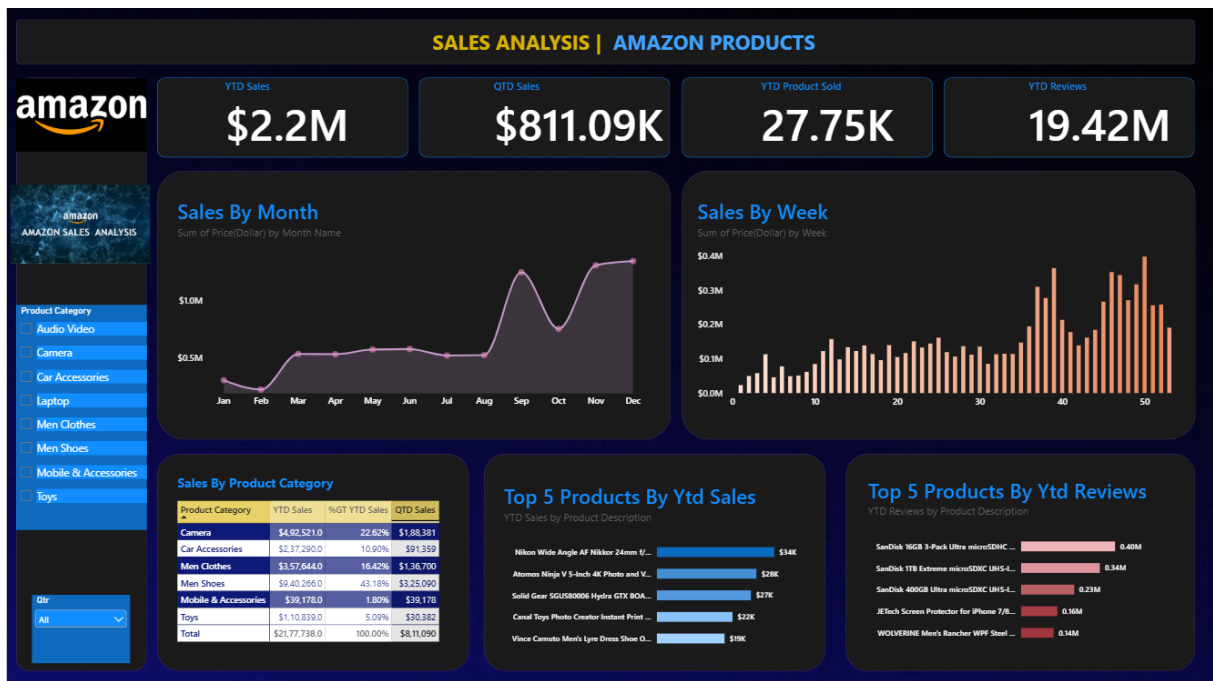


Figure 1: Amazon Sales Analysis — Power BI Dashboard

7. Key Metrics & Results

7.1 Headline KPIs

KPI	Value	Insight
YTD Sales	\$2.2M	Strong overall annual revenue performance
QTD Sales	\$811.09K	Healthy current-quarter momentum
YTD Products Sold	27.75K	High sales volume across categories
YTD Reviews	19.42M	Very high customer engagement

7.2 Sales by Product Category

Men Shoes leads category sales at 43.18% of YTD revenue, followed by Camera (22.62%) and Men Clothes (16.42%). Mobile & Accessories contributes the least at 1.80% of YTD sales.

Product Category	YTD Sales	% of YTD Sales	QTD Sales
Camera	\$4,92,521.0	22.62%	\$1,88,381
Car Accessories	\$2,37,290.0	10.90%	\$91,359
Men Clothes	\$3,57,644.0	16.42%	\$1,36,700
Men Shoes	\$9,40,266.0	43.18%	\$3,25,090

Product Category	YTD Sales	% of YTD Sales	QTD Sales
Mobile & Accessories	\$39,178.0	1.80%	\$39,178
Toys	\$1,10,839.0	5.09%	\$30,382
Total	\$21,77,738.0	100.00%	\$8,11,090

7.3 Top Performing Products

- Top product by YTD Sales: Nikon Wide Angle AF Nikkor 24mm lens (~\$34K).
- Top product by YTD Reviews: SanDisk 16GB 3-Pack Ultra microSDHC (0.40M reviews).
- Sales trend shows a clear seasonal spike from September through November, with a dip in October before recovering into year-end.
- Weekly sales volume ramps up sharply in the back half of the year (weeks 35–50), aligning with the monthly trend.

8. Insights & Conclusion

- Revenue is heavily concentrated in a few categories — Men Shoes and Camera together account for nearly two-thirds of YTD sales, suggesting these should be prioritized for inventory and marketing focus.
- Sales are highly seasonal, peaking in Q4 (Sep–Nov), which is useful for planning promotions and stock levels ahead of the season.
- High review volume relative to units sold indicates strong customer engagement, which can be leveraged in marketing for top-reviewed products.
- The dashboard's slicers (Product Category and Quarter) allow stakeholders to self-serve drill-downs without needing a new report for every question.

Overall, this project demonstrates a complete Power BI workflow — from raw CSV ingestion and data cleaning, through DAX-driven time-intelligence measures, to a polished, decision-ready dashboard — delivering clear, actionable visibility into Amazon product sales performance.